

MOUNIKA TATINENI

Generative AI & NLP Engineer

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SUMMARY

Results-driven **AI/ML Engineer** with 3+ years of experience spanning software engineering, data analytics, and **Generative AI solutions**. Proven track record in designing, fine-tuning, and deploying **LLM-powered applications** that boost automation, accuracy, and business outcomes. Skilled in **NLP, RAG pipelines, prompt engineering, and cloud AI services** (Azure OpenAI, AWS Sage maker), with hands-on expertise in **Python, LangChain, Hugging Face, and REST API integration**. Delivered measurable impact, including **3: improvement in client engagement metrics** and **50%⁺ reduction in manual workloads** through AI-driven automation. Adept at collaborating with cross-functional teams to translate business needs into scalable AI products, ensuring alignment with enterprise goals. Passionate about leveraging cutting-edge AI technologies to deliver transformative, data-driven business solutions.

EXPERIENCE

AI/ML Engineer – Generative AI

Texas

Epsilon

02/2025 - Present

Spearheaded the successful launch of AI-powered personalization and analytics tools that improved client engagement metrics by 3, automated high-volume workflows, and positioned Epsilon as a leader in enterprise Generative AI adoption.

- Leading the integration of **Generative AI** tools into Epsilon's marketing analytics workflows, improving campaign performance analysis efficiency by **40%**.
- Designed, architected, and deployed enterprise-scale Generative AI applications leveraging LLMs (GPT-4, Claude, LLaMA) to automate marketing analytics, personalized content creation, and campaign optimization.
- Engineered Retrieval-Augmented Generation (RAG) pipelines using LangChain, Pinecone, and FAISS to deliver contextually accurate responses, improving response precision by 3: and reducing irrelevant outputs by 20%.
- Fine-tuned pre-trained language models on large-scale domain datasets to enhance targeted content generation, increasing accuracy by 25% in client-specific marketing campaigns.
- Developed and automated data preprocessing and analytics workflows in Python, enabling real-time processing of millions of customer interactions, cutting reporting timelines by 50%, and reducing manual errors by 40%.
- Integrated AI solutions into enterprise marketing platforms via REST APIs and cloud-based AI services (Azure OpenAI, AWS Sage maker), decreasing deployment cycles by 30% and accelerating time-to-market for AI features.
- Led collaborative workshops with product managers, data scientists, and engineering teams to identify high-value AI opportunities, resulting in 5= production-ready AI solutions adopted by enterprise clients.
- Conducted prompt engineering and optimization for LLMs to minimize hallucinations and improve factual accuracy, boosting trust and adoption rates among stakeholders.
- Implemented monitoring and evaluation pipelines to track model performance, ensuring compliance with client requirements and industry standards.

AI/ML Engineer – Generative AI & NLP

Hybrid

Uber

05/2024 - 12/2024

Delivered AI-powered prototypes that cut financial data processing time by 40%, enhanced decision-making accuracy by 30%, and paved the way for enterprise-wide Generative AI adoption.

- Developed Gen AI-powered tools to automate **financial reporting and audit work flows**, which reduced manual processing time by **45%** across regional finance teams.
- Designed and developed LLM-powered finance analytics solutions using GPT-4, Lang Chain, and vector

databases, reducing manual reporting effort by 40%.

- Built and optimized NLP pipelines for entity recognition, summarization, and classification, achieving 95% accuracy across datasets containing over 1 million financial records.
- Implemented Retrieval-Augmented Generation (RAG) workflows with Pinecone and FAISS, increasing response relevance and accuracy by 30%.
- Fine-tuned pre-trained LLMs on domain-specific data to improve financial compliance query accuracy by 15%.
- Integrated AI models into Uber's internal finance platforms using REST APIs, cutting model deployment and integration time by 25%.
- Collaborated with cross-functional teams to identify Generative AI use cases, delivering more than three successful prototypes that accelerated decision-making processes.

Software Engineer (AI/ML)

India

Trigent Software

01/2021-12/2022

Contributed to the successful delivery of multiple AI-driven solutions, reducing manual processing time by up to 30% and improving data pipeline efficiency for client projects.

- Developed and maintained software components in Python and Java, enabling integration of AI/ML capabilities into enterprise applications, impacting over 5 client projects.
- Built and optimized data preprocessing pipelines handling datasets of up to 2M+ records, reducing data preparation time by 35%.
- Assisted in the design, testing, and deployment of machine learning models achieving 85–90% prediction accuracy for proof-of-concept initiatives.
- Collaborated with cross-functional teams to implement REST APIs for AI model inference, cutting model integration time by 25%.
- Participated in code reviews, debugging, and optimization, improving system performance by 20% and reducing defect rates by 15%.
- Authored technical documentation covering AI workflows, APIs, and deployment processes, improving onboarding time for new engineers by 40%.

SKILLS

AI & Machine Learning: Generative AI, Natural Language Processing (NLP), Large Language Models (LLM), Deep Learning (CNNs, RNNs, LSTMs, Transformers), Retrieval-Augmented Generation (RAG), Supervised & Unsupervised Learning, Prompt Engineering, Machine Learning Algorithms, Model Fine-Tuning, Vector Databases (Pinecone, FAISS), Computer Vision, Recommendation Systems.

Programming & Frameworks: Python, Java, JavaScript, TensorFlow, PyTorch, Scikit-learn, LangChain, Hugging Face Transformers, OpenAI API, REST APIs, Spring Boot, React.js, Node.js.

Data & Analytics: Data Preprocessing, Feature Engineering, SQL, NoSQL, Pandas, NumPy, Data Visualization (Power BI, Tableau, Matplotlib, Seaborn), ETL Pipelines, BigQuery, Snowflake.

Cloud & Deployment: AWS (SageMaker, Lambda, S3, EC2), Azure AI Services, Google Cloud AI Platform, GCP AI Platform, Docker, Prometheus, Kubernetes, CI/CD Pipelines, Git, MLOps.

Domain Knowledge: Healthcare Systems, HIPAA Compliance, Enterprise SaaS Solutions, E-commerce Platforms, Finance Data Analytics, Clinical Data Analytics.

Visualization & Reporting: Power BI, Tableau (basic), Excel (Pivot Tables, Power Query), Matplotlib, Seaborn, Excel (Advanced), Interactive Dashboards.

Dashboarding & Automation: Power BI Embedded, Report Automation with Python, Data-Driven Narratives using LLMs, Jupyter Notebooks, Git

Soft & Business Skills: Data Storytelling, Technical Documentation, Cross-functional Collaboration,

Financial Analysis, Communication, Requirements Gathering, Agile/Scrum (basic), Mentorship, Problem Solving

EDUCATION

Master's degree in Computer Science
University of North Texas

Texas City, United States
01/2023 - 12/2024

Bachelor of Technology in Electrical and Electronics Engineering
Siddhartha Institute of Science and Technology

Puttur, India
06/2018 - 04/2022

CERTIFICATION

Ultimate AWS Certified Developer Associate

The Python Mega Course 2022